A Novice's Adventures in ESRI Web Map Technologies

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Student Intern

Web Mapping Novice Status: Very

Who am I?

Undergraduate>Cartography/GIS Major>Senior

- Basic knowledge of cartographic principles
- Basic ArcGIS Online (AGOL), ArcGIS Pro experience

State Cartographer's Office>Student Intern

Wisconsin Sea Grant Institute>Student Intern

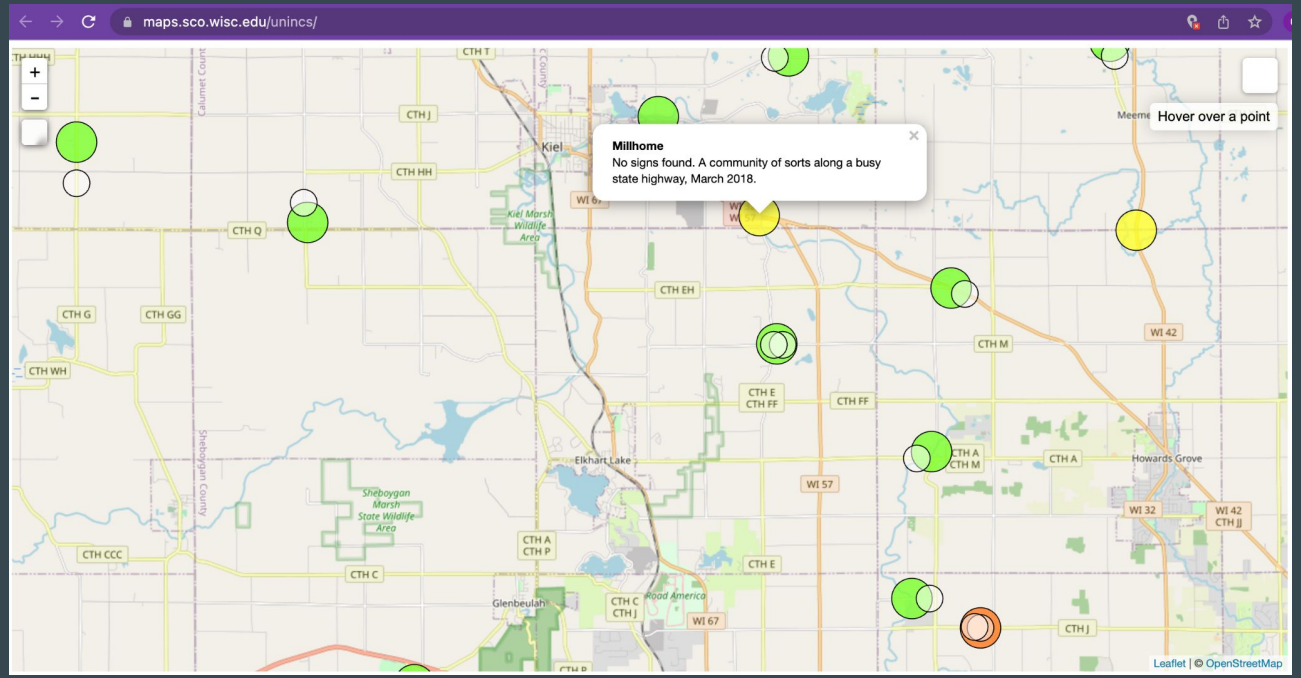
Coding experience at the beginning of the year? Zero

Comfortable with GUI interfaces, some command line

Background

We have a cool dataset of unincorporated places in Wisconsin. Attributes are ‘found’ or ‘not found’. Wanted to build on it.

Needed to choose a methodology to display the layers.



The Problem

We have a cool map or cool data and we want people to use it.

Which (ESRI) application should we use to share our creation?

- What skills do we have?
- How customizable do we need it to be?
- What kind of visual story do we need to tell?
- Who is the audience?

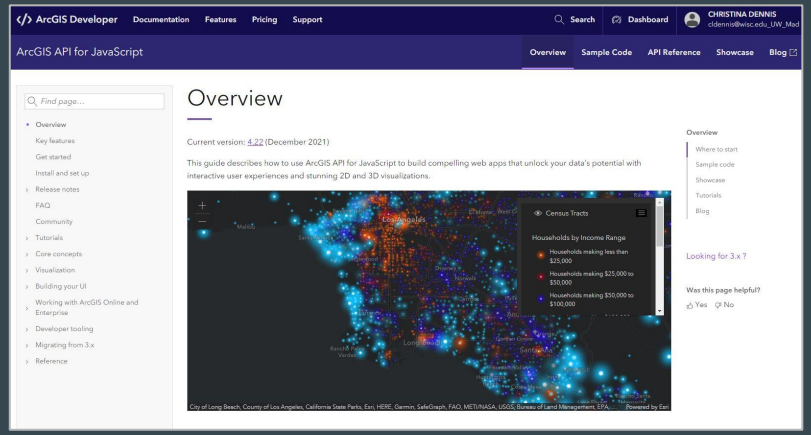
Major theme:

ArcGIS Online

All four apps (can) start with a map in AGOL

The Four Apps

1) Web AppBuilder
2) Experience Builder
3) Story Maps
4) ArcGIS API for JS

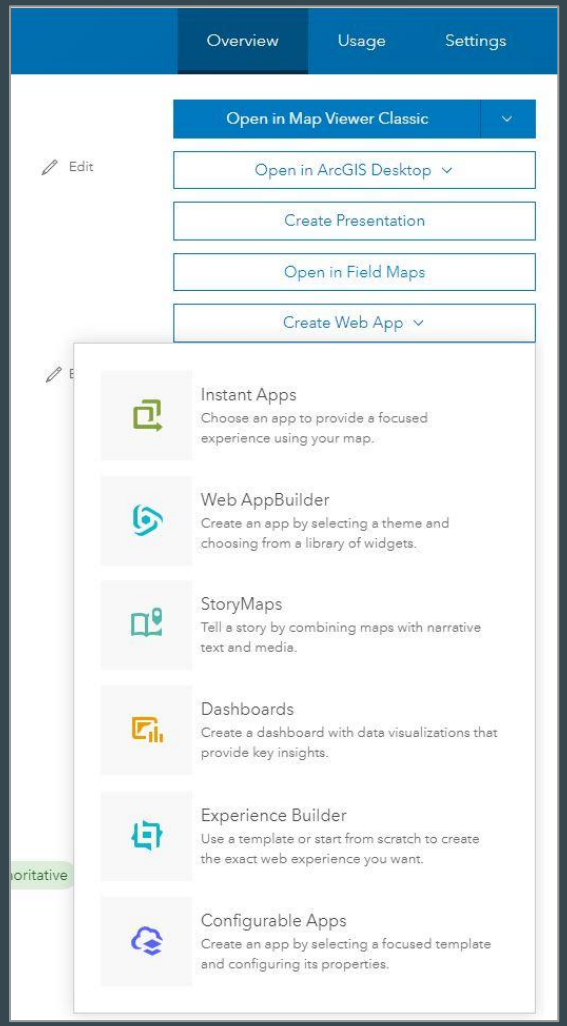


[ESRI Documentation](): A lifeline

AGOL Web Map to Web App

Upload CSV/Shapefile>Create Hosted Feature Layer>
Save as Web Map

Example

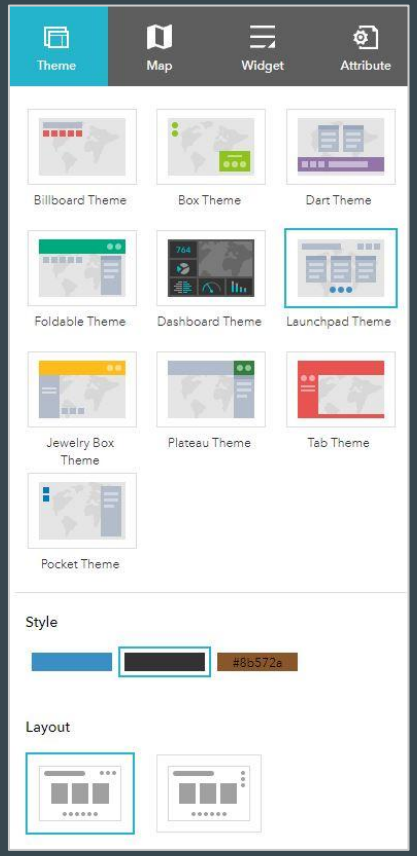


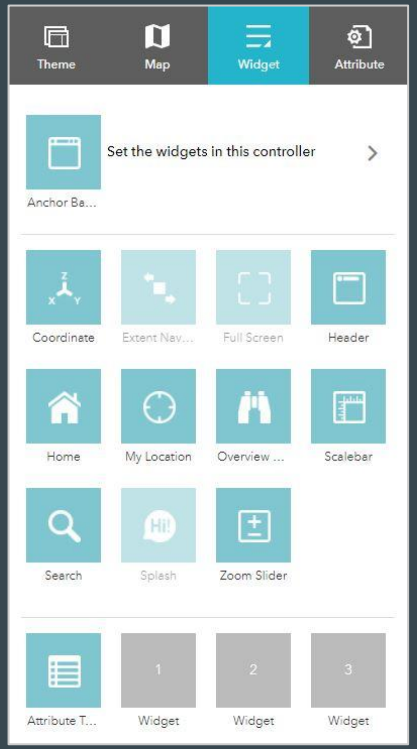
Web AppBuilder

3 Basic Steps:
1. Choose a theme
2. Add Widgets
3. Launch

Editor

App





Web AppBuilder

Summary

Skills: Basic

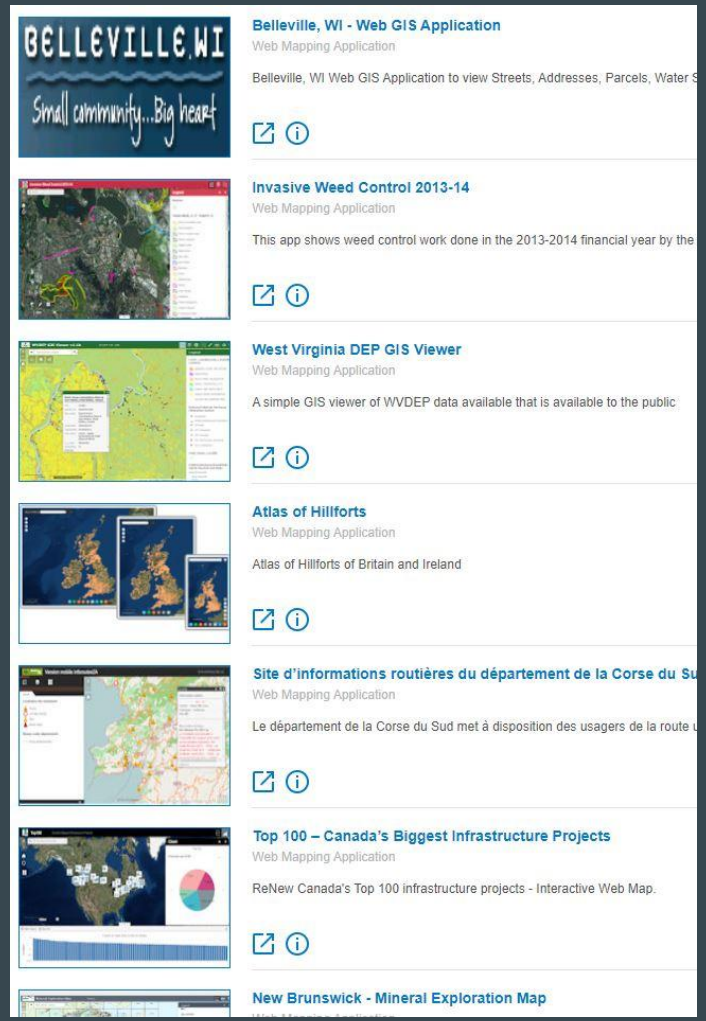
Customization: Limited (WYSIWYG)

Audience: Viewer, explorer, not much explanation

Common use: Viewer– Parcel, Property, Campus

*There is a developer version that you can customize more

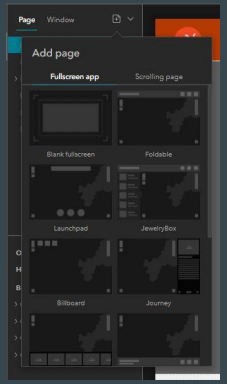
Web AppBuilder Gallery



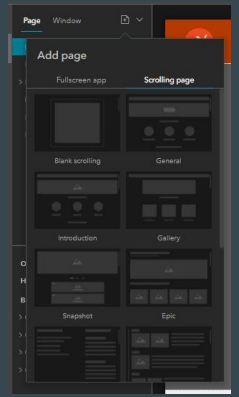
Experience Builder

Feels like a website

Start from scratch or choose a template

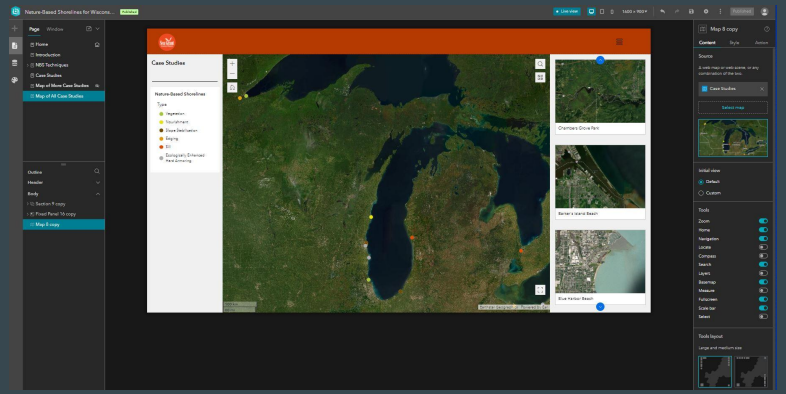


Map focused

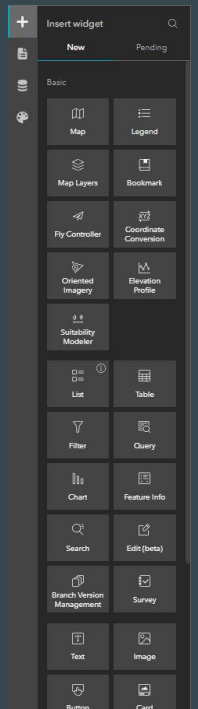


Content focused

Add maps



Add widgets



Editor App

Experience Builder

Summary

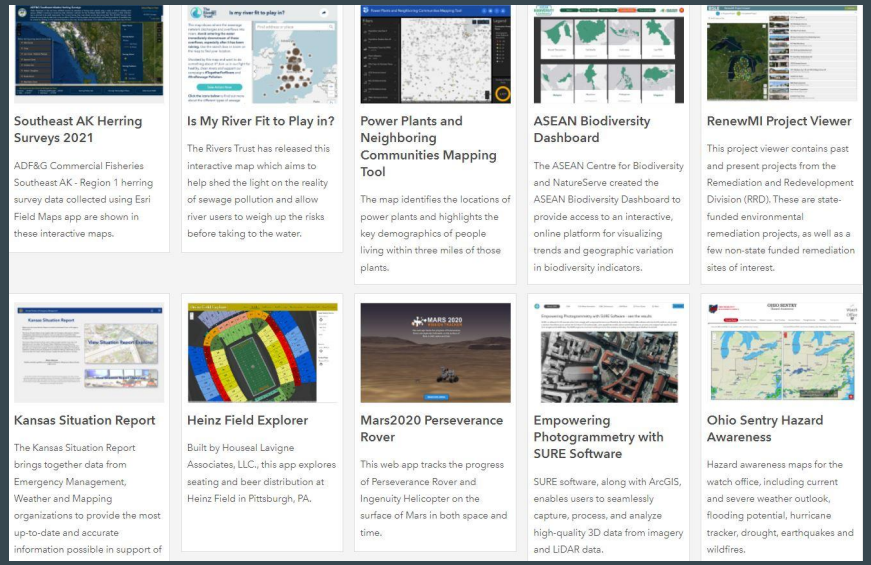
Skills: Basic

Customization: Layout, text, design, content.

Audience: Viewer, user

Common Use: Varied

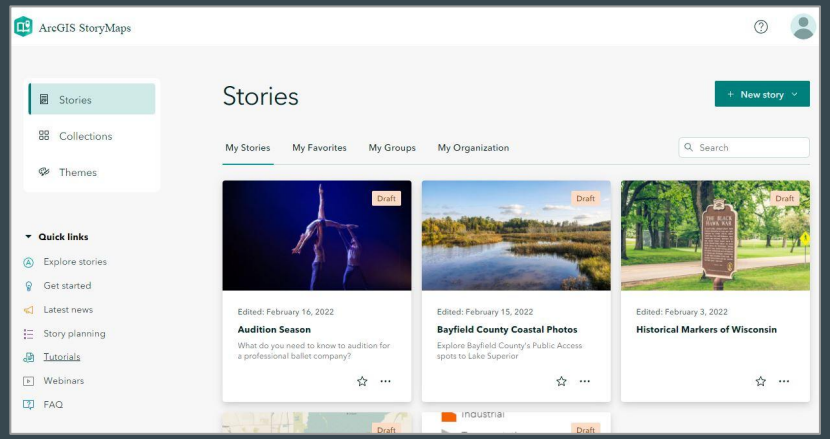
Developer Version
https://developers.arcgis.com/pricing/



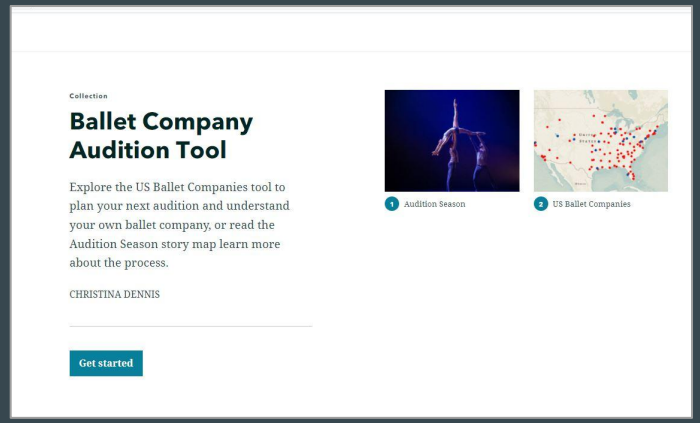
Experience Builder Gallery

Story Maps

No Coding, doesn’t support customized code.



Story



Collection

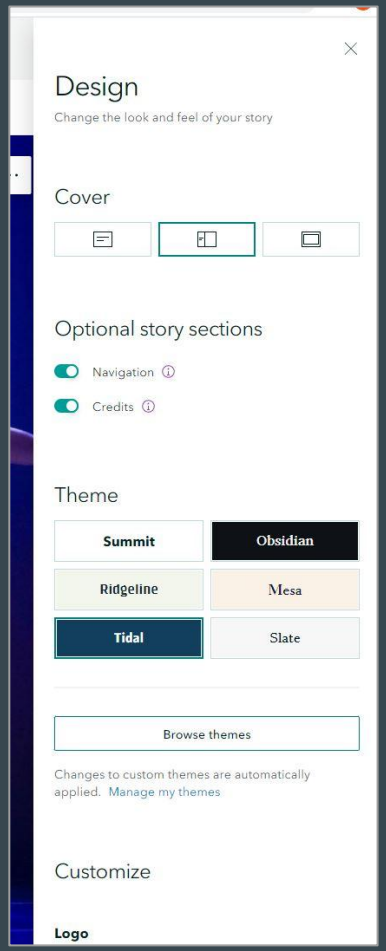
Story Maps

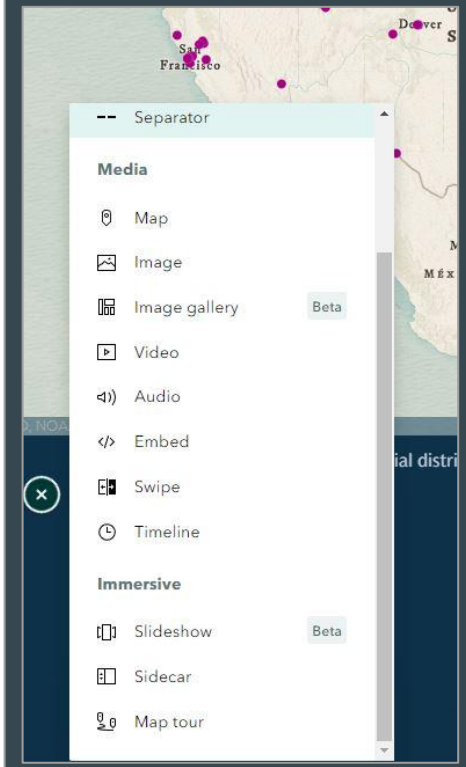
- Pick Themes
- Add Maps
- Add Narrative

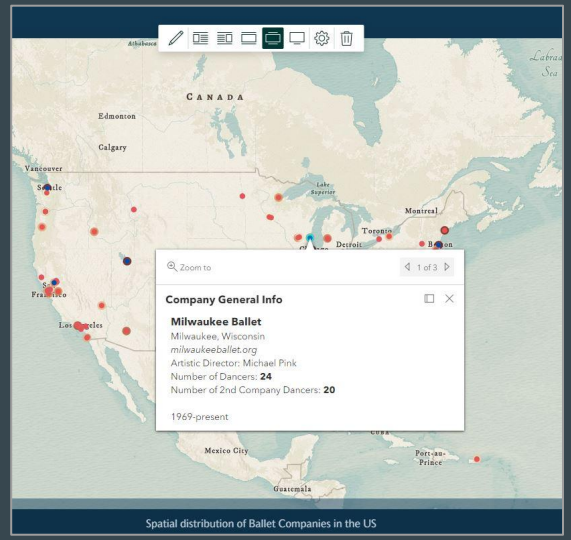
*Maps are interactive, you can choose which layers to show, but no widgets

Editor

App







Story Maps

Summary

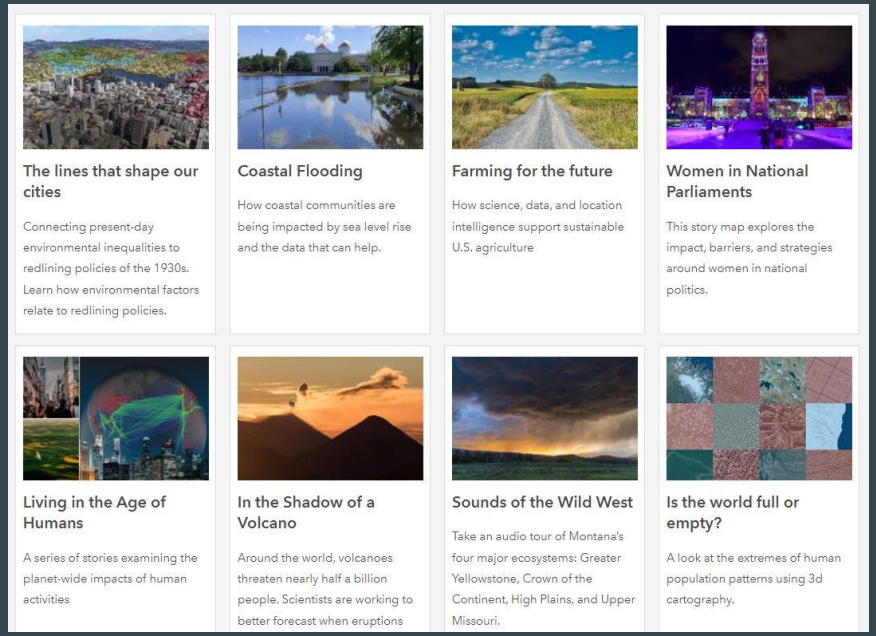
Skills: Basic

Customization: Layout, tabs, text, design, content, collections

Audience: Everyone!

Common Use: Spatial narratives, interactive story experience

Story Maps Gallery



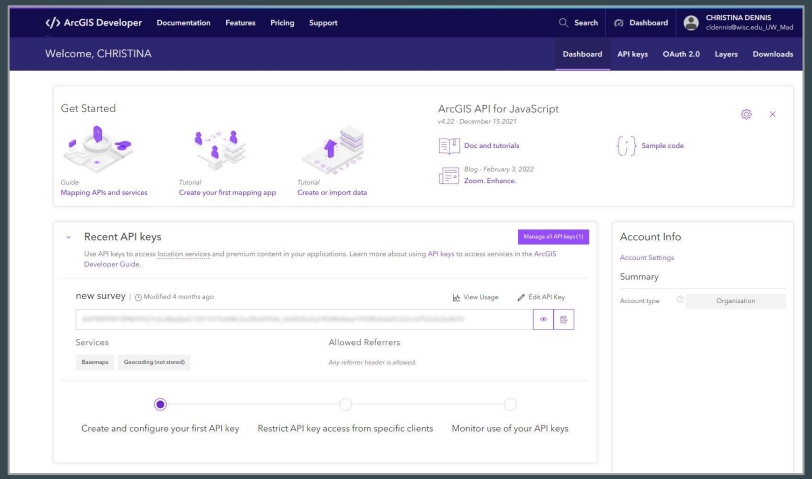
ArcGIS API for JavaScript

Coding Required

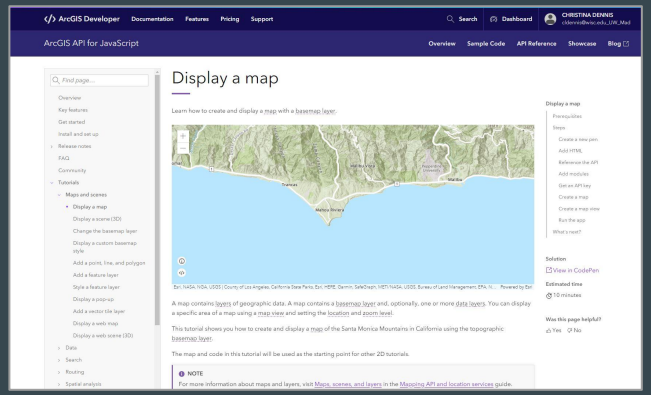
Need: Developer Dashboard–API key (AGOL comes with this)

HTML, CSS, JS knowledge

Variety of tools necessary for coding



Documentation



ArcGIS API for JS

- Integrate frameworks
  - Calcite is a bootstrapy ESRI framework
- Use ArcGIS Widgets
- Lots of tutorials and sample code
- You can connect an AGOL map as the base that you build on top of...or not
- ESRI Leaflet

Summary

Web AppBuilder

- No Coding
  - Coding with developer
- Map-Centric
- Widgets

Experience Builder

- No Coding
- Map/Narrative
- Website-like
- Widgets

Story Maps

- No Coding
- Collections
- Narrative/Map
- No widgets

API for JS

- Coding
- Dynamic
- Widgets
- Customization